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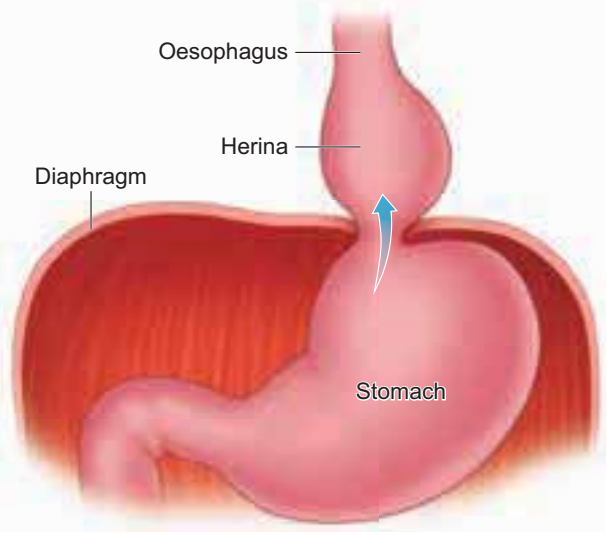


Figure 5.10 Hiatus hernia

Peptic ulcer: It refers to an eroded area of the tissue lining (mucosa) in the stomach or duodenum. Duodenal ulcer occurs in people in the age group of 25 - 45 years. Gastric ulcer is more common in persons above the age of 50 years. Ulcer is mostly due to infections caused by the bacterium ***Helicobacter pylori.*** It may also be caused due to uncontrolled usage of aspirin or certain antiinflammatory drugs. Ulcer may also be caused due to smoking, alcohol, caffeine and psychological stress.

Obesity: It is caused due to the storage of excess of body fat in adipose tissue. It may induce hypertension, atherosclerotic heart disease and diabetes. Obesity may be genetic or due to excess intake of food, endocrine and metabolic disorders. Degree of obesity is assessed by body mass index (**BMI**). A normal BMI range for adult is 19-25; above 25 is considered as obese. BMI is calculated as body weight in Kg, divided by the square of body height in meters. For example, a 50 Kg person with a height of 160 cms would have a BMI of 19.5.

That is BMI = $50/1.6^2 = 19.5$



Nobel Prize for the year 2005 was awarded to Robin Warren and Barry Marshall for the discovery of *Helicobacter pylori* which causes peptic ulcer.

Activity

Test for Starch: Add a few drops of iodine to the given warm food sample. If any starch is present in the given food sample it will change the colour of the iodine from brown to blue-black.

Test for protein: Mix the given food sample with 3mL of water in a test tube. Shake the mixture, and then add a few drops of Biuret solution. If protein is present, the colour of the solution will change to purple.

Test for glucose: Mix the given food sample with 3mL of water in a test tubes. Shake the mixture, and then add a few drops of Benedict's solution. Keep the test tube in a water bath and heat carefully. If glucose is present, the colour of the solution will change from blue to green to brick red depending upon the amount of glucose.